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The Selection of Immigrants

THE ECONOMIC IMPACT of immigration ultimately depends on the differences between the skill distributions of immigrants and natives. At any point in time, the human capital stock of the foreign-born population in a host country consists of the skills that the immigrants acquired prior to migration and of the skills acquired in the post-migration period.

This chapter examines the determinants of the volume of preexisting skills. It starts my exploration of immigration economics by asking a simple question: Who moves? Immigrants are not a randomly selected sample of the population of the source countries; some people choose to migrate and some choose to stay. The skill distribution of immigrants at the time of entry depends entirely on the nature of the selection algorithm that separates the movers from the stayers. As a result, any attempt to understand the skill differentials between immigrants and natives must begin with an analysis of the factors that motivate only some persons in a source country to emigrate.

The Migration Decision

It is instructive to begin by considering a two-country framework. Residents of a source country (country 0) consider migrating to a host country

(country 1). Assume initially that the migration decision is irreversible, so that no return migration occurs. Residents of the source country face the earnings distribution:

$$\log w_0 = \mu_0 + v_0, \tag{1.1}$$

where w_0 gives earnings in the source country; μ_0 gives the mean (log) earnings in the source country; and the random variable v_0 is i.i.d., measuring deviations from mean earnings and assumed to be normally distributed with mean zero and variance σ_0^2. For expositional convenience, equation (1.1) suppresses the subscript that indexes a particular individual.

If the *entire* population of the sending country were to emigrate, the earnings distribution that this population would face in the receiving country is:

$$\log w_1 = \mu_1 + v_1, \tag{1.2}$$

where μ_1 gives the mean (log) earnings in the host country *for this particular population,* and the random variable v_1 is also i.i.d. and normally distributed with mean zero and variance σ_1^2. The parameter ρ_{01} gives the correlation coefficient between v_0 and v_1.

As defined, the population mean μ_1 need not equal the mean earnings of native workers in the receiving country. The average worker in the sending country may be more or less skilled than the average worker in the receiving country. It is convenient to initially assume that the average person in both countries is equally skilled (or, equivalently, that any differences in average skills have been controlled for), so that μ_1 also gives the mean earnings of natives in the receiving country. This assumption helps isolate the impact of the selection process on the skill composition of the immigrant population and provides a simple way to compare the skills of immigrants and natives in the receiving country.

Equations (1.1) and (1.2) describe the earnings opportunities available to persons residing in the source country. The insight that migration decisions are motivated mainly by wage differentials can be attributed to Sir John Hicks. In *The Theory of Wages,* Hicks (1932, p. 76) argued that "differences in net economic advantages, chiefly differences in wages, are the main causes of migration." Practically all studies of migration decisions use this conjecture as a point of departure. Suppose then that the migration decision is determined by a comparison of earnings opportunities across countries, net of migration costs.[1] Define the index function:

$$I = \log\left(\frac{w_1}{w_0 + C}\right) \approx (\mu_1 - \mu_0 - \pi) + (v_1 - v_0), \tag{1.3}$$

where C gives the level of migration costs, and π gives a "time-equivalent" measure of these costs ($\pi = C/w_0$). A person emigrates if $I > 0$, and remains in the source country otherwise.

Migration is costly, and these costs likely vary among persons. However, the nature of the correlation between costs (whether in absolute dollars or in time-equivalent terms) and skills is unclear. Migration costs involve direct expenditures (such as transportation costs), forgone earnings (such as the opportunity cost of a post-migration unemployment spell), and psychic costs (such as the disutility associated with leaving family and friends behind). Suppose that the distribution of the random variable π in the sending country's population is given by:

$$\pi = \mu_\pi + v_\pi, \tag{1.4}$$

where μ_π is the mean level of migration costs in the population, and the random variable v_π is i.i.d., and assumed to be normally distributed with mean zero and variance σ_π^2. The correlation coefficients between v_π and the random variables v_0 and v_1 are given by $\rho_{\pi 0}$ and $\rho_{\pi 1}$, respectively. Given the diverse components that enter the calculation of C, it is difficult to make any sensible *a priori* inferences about the relationship between time-equivalent migration costs and skills. The specification of migration costs in equation (1.4) implies that we can rewrite the index function in (1.3) as:

$$I = \log\left(\frac{w_1}{w_0 + C}\right) = (\mu_1 - \mu_0 - \mu_\pi) + (v_1 - v_0 - v_\pi). \tag{1.5}$$

The probability that a person migrates to the host country is given by:

$$P(z) = \Pr[I > 0] = \Pr[v > -(\mu_1 - \mu_0 - \mu_\pi)] = 1 - \Phi(z), \tag{1.6}$$

where $v = v_1 - v_0 - v_\pi$; $z = -(\mu_1 - \mu_0 - \mu_\pi)/\sigma_v$; and Φ is the standard normal distribution function. Equation (1.6) summarizes the economic content of the Hicksian theory of migration. In particular:

$$\frac{\partial P}{\partial \mu_0} < 0, \ \frac{\partial P}{\partial \mu_1} > 0, \text{ and } \frac{\partial P}{\partial \mu_\pi} < 0. \tag{1.7}$$

The migration rate falls when the mean income in the source country rises, when the mean income in the host country falls, and when (mean) time-equivalent migration costs rise. Most studies in the literature on the internal migration of persons within a particular country focus on testing these theoretical predictions. The evidence reported in the internal migration literature is generally supportive of the implications in equation (1.7).[2]

Why Are There So Few Immigrants?

There exist huge income differences across countries. The Penn World Table, for example, reports that the (PPP-adjusted) per capita GDP in 2010 ranged from $41,400 in the United States to $34,900 in Ireland, $10,400 in Turkey, and $1,100 in Uganda (Heston et al. 2012). These income gaps should generate very large flows of international migrants. In the absence of migration costs, these wage differences are sufficiently large that many countries should have "emptied out" by now if their populations were composed exclusively of rational, income-maximizing agents. Of course, most receiving countries enact restrictive immigration policies that prevent these flows from occurring, and the income differences persist.

It is easy to show, however, that the existence of restrictive immigration policies cannot be the sole factor explaining why many individuals do not move across countries in search of higher income opportunities. Even when there are no legal restrictions preventing entry into high-income regions, many people in low-income countries *choose* not to move. This type of behavior is consistent with the income-maximization model of migration only if migration costs C are very high. In fact, it is easy to apply the framework to get a back-of-the-envelope estimate of the magnitude of these costs.

Consider, for example, the existing wage differences between Puerto Rico and the United States. In 2012 the mean annual earnings of workers employed in production occupations was $22,600 in Puerto Rico and $34,500 in the United States (U.S. Bureau of Labor Statistics 2012).[3] Because Puerto Ricans are U.S. citizens by birth, there are no legal restrictions preventing their move to the United States. In fact, the large income gap has induced about a third of the Puerto Rican population to migrate to the United States since the end of World War II (Borjas 2008). But, just as important, two-thirds of Puerto Ricans chose not to move.

Let w_{PR} be the wage a production worker can earn in Puerto Rico, and w_{US} be the wage he can earn in the United States. For simplicity, assume these wages are constant over a worker's career and that workers live forever. The discounted present value in each location for the typical worker is then given by:

$$PV_{PR} = \frac{w_{PR}}{r} \quad \text{and} \quad PV_{US} = \frac{w_{US}}{r}, \tag{1.8}$$

where r is the rate of discount. The *marginal* Puerto Rican production worker is indifferent between moving and staying if the discounted gains from moving are exactly equal to migration costs:

$$\frac{(w_{US} - w_{PR})}{r} = C^* \tag{1.9}$$

where C^* gives the level of migration costs for the marginal worker. Let $\pi^* = C^*/w_{PR}$, so that π^* gives the costs of migration in time-equivalent terms. I can then rewrite equation (1.9) as:

$$\frac{1}{r}\frac{(w_{US} - w_{PR})}{w_{PR}} = \pi^* \tag{1.10}$$

The ratio $(w_{US} - w_{PR})/w_{PR}$ is around 0.5, indicating that a "typical" Puerto Rican production worker can increase his income by 50 percent by migrating to the United States. If the rate of discount is 5 percent, the left-hand side of equation (1.10) takes on the value of 10. In other words, migration costs for the marginal worker who is indifferent between migrating to the United States and staying in Puerto Rico are 10 times his salary. If this production worker earns the average salary in Puerto Rico, migration costs are around $226,000![4]

What exactly is the nature of these costs? They obviously do not represent the cost of transporting the family and household goods to a new location in the United States. Instead, the marginal Puerto Rican must attach a very high utility to the social and cultural amenities associated with remaining in his birthplace. Needless to say, migration costs could be even larger in other contexts—where there are much greater differences in language and culture between sending and receiving locations, and where the receiving country does not have a large and thriving network of compatriots to welcome and ease the entry of new migrants.[5] In short, the Puerto Rican example dramatically suggests that many people would choose not

to migrate even if high-income receiving countries did not impose legal impediments.[6]

The Roy Model

Although it is important to determine the size and direction of migration flows, it is equally important to determine *which* persons find it most worthwhile to migrate to the receiving country. Even in the absence of legal restrictions impeding international migration, only a subset of persons in the host country chooses to move. The nature of the selection that generates a nonrandom sample of immigrants is closely related to the Roy model of occupational choice (Roy 1951). Roy's original setting examined a worker's choice between becoming a fisherman or a hunter, and provided an algorithm that described which types of people would go into each of the two occupations. The framework was first applied to answer the question "Should I stay or should I go?" in Borjas (1987b, 1991).[7]

Consider the conditional expectations $E(\log w_0 \mid I>0)$ and $E(\log w_1 \mid I>0)$. These expectations give the average earnings in both the source and the host country for the self-selected sample of migrants. It is important to emphasize that these expectations hold the parameters $(\mu_0, \mu_1, \sigma_0^2, \sigma_1^2, \rho_{01})$ of the income distributions constant. This is not a trivial assumption. For example, the model explicitly assumes that the migration flow is sufficiently small so that there are no feedback effects on the average income of immigrants (or natives) in the host country or on the average income of the "stayers" in the source country. A general equilibrium model would account for the fact that the parameters of the income distributions in both sending and receiving regions are endogenous, and likely depend on the size and skill composition of the immigrant flow.

Because the random variables v_0, v_1, and v_π are jointly normally distributed, it can be shown that these conditional means are given by:[8]

$$E(\log w_0 \mid I>0) = \mu_0 + \frac{\sigma_0 \sigma_1}{\sigma_v}\left[\left(\rho_{01} - \frac{\sigma_0}{\sigma_1}\right) - \rho_{\pi 0}\frac{\sigma_\pi}{\sigma_1}\right]\lambda(z), \tag{1.11}$$

$$E(\log w_1 \mid I>0) = \mu_1 + \frac{\sigma_0 \sigma_1}{\sigma_v}\left[\left(\frac{\sigma_1}{\sigma_0} - \rho_{01}\right) - \rho_{\pi 1}\frac{\sigma_\pi}{\sigma_0}\right]\lambda(z), \tag{1.12}$$

where $\lambda(z) = \phi(z)/(1 - \Phi(z)) \geq 0$, and ϕ is the density function of the standard normal distribution.

It is easier to grasp the economic intuition behind the implied selection mechanism by initially assuming that $\sigma_\pi = 0$, so that time-equivalent migration costs are constant in the population. To simplify the discussion, let $Q_0 = E(v_0 \mid I > 0)$ and $Q_1 = E(v_1 \mid I > 0)$. Manipulation of equations (1.11) and (1.12) shows that the Roy model demarcates three cases that summarize the pattern of skill differences between immigrants and natives:[9]

$$\begin{aligned} &Q_0 > 0 \text{ and } Q_1 > 0, \quad \text{if} \quad \rho_{01} > \frac{\sigma_0}{\sigma_1} \text{ and } \frac{\sigma_0}{\sigma_1} < 1, \\ &Q_0 < 0 \text{ and } Q_1 < 0, \quad \text{if} \quad \rho_{01} > \frac{\sigma_1}{\sigma_0} \text{ and } \frac{\sigma_0}{\sigma_1} > 1, \\ &Q_0 < 0 \text{ and } Q_1 > 0, \quad \text{if} \quad \rho_{01} < \min\left(\frac{\sigma_1}{\sigma_0}, \frac{\sigma_0}{\sigma_1}\right). \end{aligned} \tag{1.13}$$

Define *positive selection* to occur when immigrants have above-average earnings in both the source and the host country ($Q_0 > 0$ and $Q_1 > 0$), and *negative selection* when immigrants have below-average earnings in both countries ($Q_0 < 0$ and $Q_1 < 0$). Equation (1.13) shows that either type of selection requires that skills be (sufficiently) positively correlated between the two countries. The variances σ_0^2 and σ_1^2 measure the "price" of skills: the greater the rewards to skills, the larger the inequality in wages.[10] Immigrants are then positively selected when the sending country—*relative to the receiving country*—offers a lower return to skills, "taxing" high-skill workers and "insuring" low-skill workers against poor labor market outcomes. In contrast, immigrants are negatively selected when the receiving country taxes high-skill workers and subsidizes low-skill workers.

The Roy model reveals that there exists a final selection possibility: the receiving country draws persons who have below-average earnings in the source country but do well in the host country ($Q_0 < 0$ and $Q_1 > 0$). This *inverse sorting* occurs when the correlation coefficient ρ_{01} is small or negative. The correlation could be negative when, for example, a source country experiences a political upheaval that results in the takeover of the government by a very redistributive regime, such as a communist revolution. In its initial stages, this political system often redistributes incomes by confiscating the assets of relatively successful persons. Immigrants from such systems will

be in the lower tail of the post-revolution income distribution, but they will have the skills and wherewithal to perform quite well in the host country's market economy.

Equation (1.13) shows that neither differences in mean incomes across countries nor the mean level of migration costs determines the *type* of selection that characterizes the immigrant flow. Mean incomes and migration costs affect the size of the flow (and hence the extent to which the skills of the average immigrant differ from the mean skills of the base population) through the variable $\lambda(z)$, but they do not determine whether the immigrants are drawn mainly from the upper or lower tail of the skill distribution. Put differently, the gap in mean incomes and the level of migration costs influence the "intensity" of selection, but not the type.

It is important to examine what happens to the selection rules summarized in equation (1.13) if migration costs π are not constant across workers. It is easy to verify that variable migration costs do not alter any of the selection rules if (a) time-equivalent migration costs are uncorrelated with skills ($\rho_{\pi 0} = \rho_{\pi 1} = 0$), or (b) the ratio of variances σ_π / σ_j ($j = 0, 1$) is "small." In every other case, however, variable migration costs can change the nature of selection. Suppose, for example, that π is negatively correlated with skills, perhaps because high-skill workers are more proficient at collecting information about opportunities abroad or find it easier to adapt in a new environment. This negative correlation increases the likelihood that the bracketed term in equations (1.11) and (1.12) is positive, and the immigrant flow is more likely to be positively selected. Conversely, the likelihood of negative selection increases if time-equivalent migration costs and skills are positively correlated.

Inspection of equation (1.11) and (1.12) clarifies that it is not the heterogeneity in migration costs *per se* that can alter the nature of the selection, but the possibility that there exists a nonzero correlation between skills and migration costs. I would argue that the presence of such a correlation greatly reduces the empirical usefulness of the Roy model as a framework for thinking about the type of selection that characterizes the immigrant population. Put bluntly, *any* type of selection observed empirically can be explained by making the appropriate assumption about $\rho_{\pi 0}$ and $\rho_{\pi 1}$. This is particularly problematic because, as noted earlier, there is little reason to believe that we can figure out the nature of the distribution in migration costs by introspection.

For example, suppose that the pool of migrants from Mexico to the United States is composed of workers in the middle of the Mexican skill distribution (as reported in some of the empirical studies discussed below). The simplest version of the Roy model implies that Mexican migrants should be negatively selected because the rate of return to skills in Mexico is far higher than that found in the United States. Nevertheless, we can easily remedy the discrepancy between the data and the theory by assuming that migration costs are very high for the least-skilled Mexicans. These high migration costs remove the least-skilled workers from the sample of movers, leaving only workers in the middle of the skill distribution in the self-selected pool. The reverse-engineering of the "right" set of assumptions about the distribution of costs implies that the Roy model is no longer empirically falsifiable.

It is also of interest to generalize the selection rules summarized in (1.13) by examining the implications of the Roy model not only for the means of the income distributions among the relevant groups (movers and stayers), but also for the shape of the *entire* distributions. Consider the income distributions of two different populations, and let $\mathbf{F}(x)$ and $\mathbf{G}(x)$ represent the respective (cumulative) probability distributions, where x denotes the level of income for a particular person. By definition, $\mathbf{F}(x)$ stochastically dominates $\mathbf{G}(x)$ if:

$$\mathbf{F}(x) < \mathbf{G}(x), \forall\, x. \tag{1.14}$$

Stochastic dominance implies that a larger fraction of the workforce has incomes *above x* in the population represented by the dominating **F** distribution than by the **G** distribution. In other words, for any level of income, the population described by the probability distribution **F** is wealthier, because a larger fraction of the group exceeds that threshold. Note that stochastic dominance must imply a ranking of means where $\mu_F > \mu_G$, but the converse need not hold.

The selection that characterizes the sample of immigrants in the Roy model can be easily described in terms of the concept of stochastic dominance in the special case where $\sigma_\pi = 0$. In particular, let $\mathbf{F_M}(v_0)$ be the distribution function of (pre-migration) earnings in the source country for the sample of movers, and let $\mathbf{F_S}(v_0)$ be the respective distribution function for stayers. In short, $\mathbf{F_M}(v_0)$ and $\mathbf{F_S}(v_0)$ summarize the truncated earnings distributions—*prior to migration*—for the self-selected samples of movers and stayers, respectively. As shown in Appendix A, the Roy

model predicts a unique stochastic ranking of these two truncated distributions. In particular:[11]

$$\begin{aligned} \mathbf{F_M}(v_0) &< \mathbf{F_S}(v_0), \quad \text{if} \quad \frac{\sigma_0}{\sigma_1} < \rho_{01}, \\ \mathbf{F_M}(v_0) &> \mathbf{F_S}(v_0), \quad \text{if} \quad \frac{\sigma_0}{\sigma_1} > \rho_{01}. \end{aligned} \tag{1.15}$$

Suppose that the rate of return to skills in the source country is sufficiently low, so that the ratio σ_0/σ_1 is below ρ_{01}. Equation (1.11) implies that the subsample of movers will necessarily have a higher mean income prior to moving than the stayers (so that the movers are positively selected in terms of mean incomes). Equation (1.15) further states that the distribution function of pre-migration incomes of movers stochastically dominates that of stayers, so that movers are "wealthier" than stayers at any point in the income distribution.

In contrast, suppose that the rate of return to skills is higher in the source country, implying that σ_0/σ_1 exceeds ρ_{01}. This situation might arise either because immigrants are negatively selected in the sense defined by equation (1.13), or because there is an inverse sorting of immigrants and the correlation coefficient ρ_{01} is small or negative. In either case, the pre-migration mean earnings of the migrants are lower than the pre-migration mean earnings of the stayers. Equation (1.15) shows that, for any level of income, a larger fraction of stayers is "wealthier."

Let me conclude by emphasizing that although this framework has become the canonical model of selection in the immigration literature, there is an important sense in which it is incomplete and could lead to misleading predictions about the skill composition of the immigrant population. As noted above, the Roy model provides a partial equilibrium framework for identifying the types of workers who find it most worthwhile to move. The immigrant flow is assumed to be "small," so that the impact of the labor supply shifts on the wage structures of both the source and the host country is negligible and can be ignored by the various agents in the model. In fact, the international movement of large numbers of workers will likely affect both wage structures, altering the optimal size and skill composition of the immigrant influx. However, the implications of a general equilibrium framework for the shape of the wage structures and for the selection predictions of the Roy model remain unexplored.[12]

A Special Case of the Roy Model and Extensions

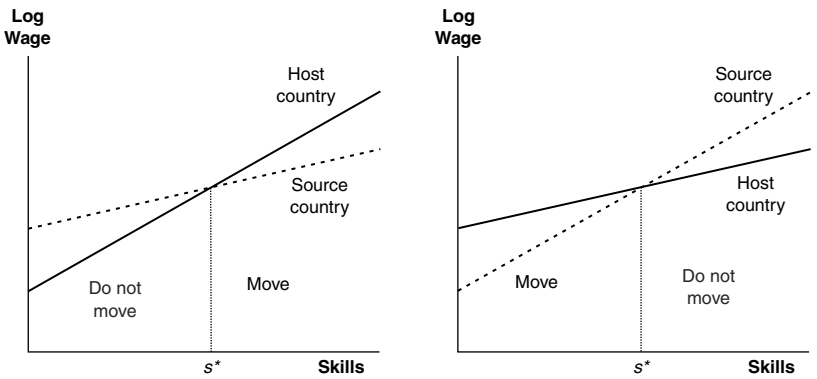
There is a special case of the Roy model that can be illustrated graphically and that provides a simple point of departure for more complex situations. In particular, suppose that the correlation coefficient ρ_{01} is equal to one. This assumption effectively implies that earnings in both the source country and the United States depend on a single factor that is perfectly transferable across countries. In particular, let s denote the number of efficiency units embodied in the worker. The assumption that ρ_{01} is equal to one implies that we can write the income distributions in the two countries as:

$$\log w_0 = \alpha_0 + r_0 s, \tag{1.16}$$

$$\log w_1 = \alpha_1 + r_1 s, \tag{1.17}$$

where r_0 and r_1 are the rate of return to skills in the two countries, respectively.

The two panels of Figure 1.1 illustrate the linear relation between a worker's log wage and skills for each of the countries. In Figure 1.1A, the wage–skills profile is steeper in the host country, so the rate of return to skills is higher in the host country, while in Figure 1.1B, the rate of return to skills is higher in the source country. To easily illustrate the type of immigrant selection implied by this simplification, assume initially that workers do not incur any costs when they move. The decision rule determining im-



A. Positive Selection **B. Negative Selection**

Figure 1.1. The Roy Model, Assuming Incomes Are Perfectly Correlated across Countries

migration is then quite simple: Workers originating in the source country emigrate whenever earnings are higher in the receiving country.[13]

Consider first the selection that occurs in Figure 1.1A. Workers with fewer than s^* efficiency units earn more if they stay in the source country than if they move. Workers with more than s^* efficiency units, however, earn more by moving. As long as the rate of return to skills is higher in the host than in the source country, therefore, the migration flow consists of workers in the upper tail of the skill distribution and immigrants are positively selected. Consider now Figure 1.1B. Workers with fewer than s^* efficiency units earn more in the host country and will want to move. When the returns to skills in the receiving country are relatively low, therefore, the immigrant flow is composed of the least-skilled workers in the source country and immigrants are negatively selected.

As Figure 1.1 shows, the assumption that ρ_{01} is equal to one leads to a simple separation of the pools of movers and stayers. Further, the graphical analysis can be easily extended to incorporate migration costs by simply shifting the respective wage profile accordingly. For example, the presence of constant time-equivalent migration costs would shift down the wage–skills profile in the host country (its intercept would now be given by $\alpha_1 - \pi$). As before, the introduction of constant migration costs does not change the nature of the selection (the movers would still be either positively or negatively selected), but would change its intensity because it would change the number of migrants.

The assumption that ρ_{01} is equal to one is useful because it permits several interesting and tractable generalizations. For example, consider the extension of the Roy model from a two-country to a multicountry framework, so that an individual residing in a particular source country must decide not only whether to leave that country, but also which potential destination to move to.

To simplify the exposition, assume that migration costs are equal to zero. A worker residing in a particular "base" country compares his earnings opportunities in all potential countries of residence and moves to the location that maximizes earnings. We can write the earnings distributions that workers in the base country would face if they all migrated to each of the n potential countries as:[14]

$$\log w_k = \alpha_k + r_k s, \qquad k = 1, \ldots, n. \tag{1.18}$$

An income-maximizing individual chooses to reside in country j whenever:

$$\log w_j > \max_{k \neq j}(\log w_k). \tag{1.19}$$

For convenience, rank countries such that $r_1 < r_2 < \ldots < r_n$. Figure 1.2 illustrates the nature of the sorting that takes place when $n = 3$. The least-skilled workers settle in the country with the lowest rate of return to skills, whereas the most-skilled workers settle in the country with the highest rate of return. In effect, income-maximizing behavior creates a positive correlation between the average skills of a country's inhabitants and the country's rate of return to skills:[15]

$$E(s \mid \text{choose } j) > E(s \mid \text{choose } k) \text{ if and only if } r_j > r_k. \tag{1.20}$$

The assumption that earnings are perfectly correlated across countries implies that individuals who rank highly in the income distribution in one country will also rank highly in the income distribution of any other country. Skilled workers, therefore, are attracted to countries that offer a relatively higher rate of return to skills. In contrast, low-skill workers choose countries that effectively "subsidize" their lack of human capital. As with the two-country Roy model, international differences in the returns to skills play an important allocative role in the sorting equilibrium.[16]

Another interesting extension, developed in Borjas and Bratsberg (1996), allows for the possibility of return migration. As in Rosen's (1972)

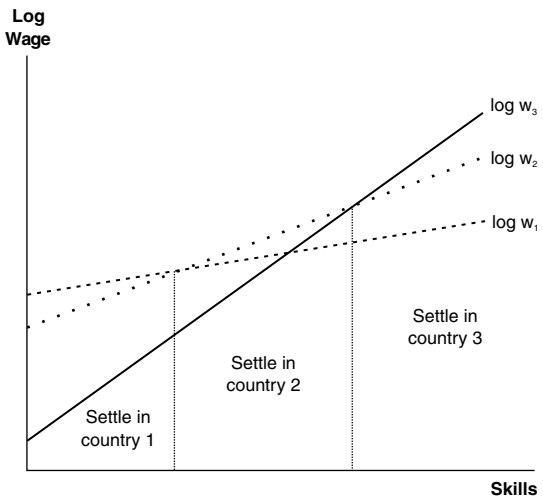


Figure 1.2. Selection in a Roy Model with Multiple Destinations

analysis of occupational mobility, return migration can occur as part of a "stepping stone" career path; it may be profitable to live abroad for a few years and acquire particular types of human capital that are valued in the source country's labor market. Return migration flows may also arise as workers correct mistakes in the initial migration decision (due perhaps to uncertainty in host country conditions). The Roy model helps to describe the type of selection that characterizes the return migrants.

To simplify the presentation, I illustrate the economic forces that determine return migration in the context of forward-looking workers who view temporary migration as a human capital investment. In a two-country framework, the log earnings distributions are given by equations (1.16) and (1.17). The worker also considers a third choice: migrating to country 1 for a fraction θ of his work life and returning home with an increased productivity. As a first-order approximation and ignoring discounting, the log wage associated with this choice is:

$$\log w_{10} = \theta \log w_1 + (1-\theta)(\log w_0 + \kappa), \tag{1.21}$$

where $0 < \theta < 1$, and κ is the rate of return to temporary migration.

Suppose that the (time-equivalent) costs of migration and return migration equal π_{01} and π_{10}, respectively. These costs, as well as the parameters θ and κ, are assumed to be constant. A worker becomes an immigrant if:

$$\log w_1 - \pi_{01} > \log w_0 \quad or \quad \log w_{10} - \pi_{01} - \pi_{10} > \log w_0. \tag{1.22}$$

In other words, a worker migrates if either the returns to permanently moving to the host country or the returns from a temporary sojourn surpass initial conditions (all net of migration costs). Similarly, a worker becomes a return migrant if:

$$\log w_{10} - \pi_{01} - \pi_{10} > \log w_0 \ \ and \ \ \log w_{10} - \pi_{01} - \pi_{10} > \log w_1 - \pi_{01}. \tag{1.23}$$

The return migrants consist of the subsample of immigrants for whom the returns from temporary migration (net of migration costs) exceed the returns both of never moving and of moving permanently to the host country.

It is easy to characterize the various groups in terms of the distribution of efficiency units s in the population. In order for there to be *any* return migration, it must be the case that κ is sufficiently large. It is easy to show that the inequalities in equation (1.23) imply:

$$\kappa > \pi_{01} + \frac{\pi_{10}}{1-\theta}, \tag{1.24}$$

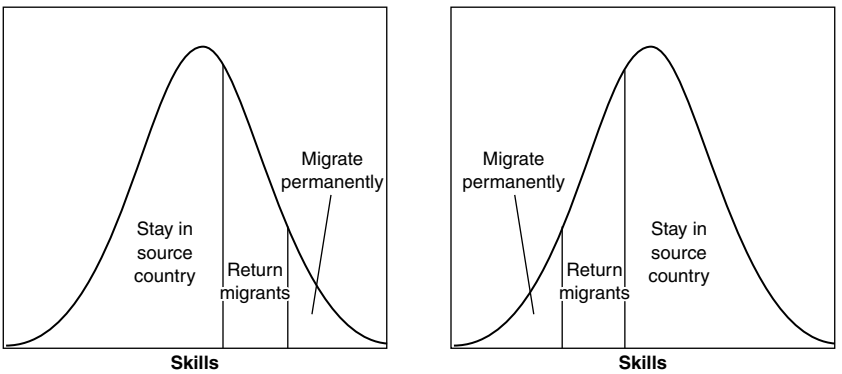
so that the rate of return to temporary migration must exceed the "expected" costs of migration. Assuming equation (1.24) holds, the subset of workers who become immigrants is given by:

$$(r_1 - r_0)s > (\alpha_0 - \alpha_1) + \kappa + \frac{\pi_{01} + \pi_{10} - \kappa}{\theta}, \tag{1.25}$$

and the subset of immigrants who become return migrants satisfies the inequality:

$$(\alpha_0 - \alpha_1) + \kappa + \frac{\pi_{01} + \pi_{10} - \kappa}{\theta} < (r_1 - r_0)s < (\alpha_0 - \alpha_1) + \kappa - \frac{\pi_{10}}{1-\theta}. \tag{1.26}$$

The two panels of Figure 1.3 illustrate the nature of selection in this model. The intuition for this sorting is obvious. Suppose, for example, that the immigrants were positively selected. The marginal immigrant in this self-selected sample is the least-skilled worker of a high-skill flow. The most-skilled workers in this sample have the most incentive to remain in the host country even if their economic opportunities improve in the source country. The marginal immigrant is most responsive to improving conditions in the



A. Positive Selection **B. Negative Selection**

Figure 1.3. Distribution of Skills and Selection in a Roy Model with Return Migration

source country and is most likely to become the return migrant. In contrast, if immigrants were negatively selected, the marginal immigrant is now the most-skilled person in a low-skill flow. Any improvements in source country conditions will particularly influence the return migration decision of the marginal immigrant, who happens to be the most-skilled worker.

In short, the return migration decision *reinforces* the original selection that characterized the immigrant population. Because it is the marginal immigrants who are most likely to become return migrants, the stayers are the "best of the best" if there is positive selection and the "worst of the worst" if there is negative selection.[17]

Linear Utility and "Generalized" Positive Selection

Although the Roy model has become a widely used framework for thinking about selection in the immigration context, its prediction that there exist plausible conditions under which the population of immigrants may be negatively selected has been disputed. In particular, it has been noted that many, if not most, immigrant flows are composed of workers who tend to have above-average education. For example, Docquier and Marfouk (2006) estimate the emigration rate from the less developed countries to be 1 percent for the general population, but over 13 percent for persons with tertiary education. These statistics suggest that the Roy model's predictive usefulness is limited, because most immigrant flows may be positively selected even though many of the source countries have very high rates of return to skills that should have generated negatively selected flows.

Grogger and Hanson (2011) propose an alternative model where positively selected flows naturally arise under most scenarios, particularly when examining migration flows from poor to rich countries. The nature of the prediction of "generalized" positive selection can be easily illustrated by examining a special case of this alternative framework. In particular, consider a situation where persons in a particular country, say 0, consider the possibility of settling in one of many potential destination countries, indexed by k. There are two skill groups in the population: high-skill (H) and low-skill (L). Finally, suppose that no other factors determine wages (so that the earnings distribution has zero variance for a particular skill group in a particular country). Let w_k^j give the wage that a worker in skill

group j ($j = H, L$) would earn in country k. The log wage offers in country k for low- and high-skill workers are given by:

$$\log w_k^L = \mu_k, \tag{1.27a}$$

$$\log w_k^H = \mu_k + \delta_k, \tag{1.27b}$$

where δ_k gives the rate of return to skills in country k.[18] Suppose also that migration costs are composed simply of a fixed cost, C_{0k}, that is incurred by all workers—regardless of skill level—when migrating from country 0 to k.

The utility of worker i if he chooses to settle in country k is assumed to be linear in dollars and can be written as:

$$U_{i0k}^j = \alpha(w_k^j - C_{0k}) + \varepsilon_{i0k}^j, \tag{1.28}$$

where $\alpha > 0$, and ε is an unobserved random variable that summarizes the worker's preferences for particular places. Worker i calculates the utility associated from settling in each of the potential destinations, as well as the utility associated with not migrating at all (where $k = 0$ and $C_{00} = 0$).

Let p_{0k}^j be the fraction of the workforce in country 0 and in skill group j that chooses to move to country k. If ε_{i0k}^j has an i.i.d. extreme value distribution, it is well known (McFadden 1974) that the log odds of migrating from country 0 to country k are given by:

$$\log \frac{p_{0k}^j}{1 - p_{0k}^j} = \alpha(w_k^j - w_0^j) - \alpha C_{0k}, \tag{1.29}$$

where $w_k^L = e^{\mu_k}$ and $w_k^H = e^{\mu_k + \delta_k}$.

It is easy to see the type of selection implied by the linear utility framework by differencing the log odds ratio in equation (1.29) between the two skill groups. The differencing yields:

$$\log \frac{p_{0k}^H}{1 - p_{0k}^H} - \log \frac{p_{0k}^L}{1 - p_{0k}^L} = \alpha[(w_k^H - w_0^H) - (w_k^L - w_0^L)]. \tag{1.30}$$

Note that the differencing eliminates the fixed migration costs because they are independent of skill.

Equation (1.30) implies positive selection occurs as long as the differenced wage gap $\left[(w_k^H - w_0^H) - (w_k^L - w_0^L)\right]$ in the right-hand side of the

equation is positive. Positive selection then requires that the *absolute* wage gap between the receiving and the source country be greater for high-skill than for low-skill workers. If migrants flow from poorer to richer countries (so that $\mu_k > \mu_0$), the differenced wage gap must necessarily be positive if $\delta_k > \delta_0$. The framework, therefore, replicates the Roy model prediction that countries offering a higher return to skills attract relatively more skilled immigrant flows.

The linear utility framework, however, also suggests that the migrant flow may be positively selected even if $\delta_k \leq \delta_0$. To see this, suppose that the rate of return to skills is the same in both countries ($\delta_k = \delta_0$). The larger absolute wage in the wealthier country ensures that $(w_k^H - w_k^L) > (w_0^H - w_0^L)$ even though the percent payoff to skills is the same in both countries.[19] In fact, if the absolute wage gap between poor and rich countries is sufficiently large, the rate of return to skills could be lower in the richer country and the immigrant flow would *still* be positively selected.

The linear utility framework has a dynamic implication that is interesting but may be troubling. Suppose that a worldwide skill-neutral technological shift continuously increases wages in all countries and for all skill groups. In the absence of migration costs, the Roy model would predict no change in the rate of international migration, because the *log* wage gap that summarizes migration incentives has not changed. The linear utility model, however, predicts either a substantial increase or a substantial decrease in the number of immigrants. This prediction follows directly from equation (1.29). Abstracting from migration costs, a steady rise in $(w_k^j - w_0^j)$ would continually increase the magnitude of the log odds ratio in the probability of moving for every skill group. If the log odds ratio were positive ($p_{0k}^j > 0.5$), the increase in world wealth would lead to exponential growth in the migration rate until it approaches 1. If the log odds ratio were negative ($p_{0k}^j < 0.5$), the migration rate would become vanishingly small. Because typically fewer than half of the people in most source countries emigrate, the linear utility model seems to suggest that the notion of international migration will become (or, perhaps, should have become) a distant memory as countries become wealthier, even though some countries will remain relatively poor.[20]

Regardless of whether this prediction of the model is empirically relevant or not, the wide array of results that can be derived from alternative selection models indicates that in the end, there will only be one way of determining which framework is most useful for understanding the nature of immigrant selection—by looking at the data.

Evidence on Selection

Cross-Country Regressions

It is well known that there is a large variance in skills and economic performance among immigrant groups residing in the United States but originating in different source countries. This dispersion across national origin groups inspired the methodology employed by the early empirical tests of the Roy model. Beginning with Borjas (1987b), a number of studies have estimated "cross-country regressions" to determine if post-migration earnings differences across national origin groups could be attributed to differences in the characteristics of the source countries. It is easy to illustrate (and update) this approach by using the Public Use Microdata Sample (PUMS) files from the 1980–2000 U.S. decennial censuses and the pooled 2009–2011 American Community Surveys (ACS). For expositional convenience, I refer to the pooled ACS data as the "2010 census." Appendix B contains a detailed description of the construction of the various data sets used throughout this book.

Each survey reports the earnings of workers belonging to specific national origin groups. To examine the differences in the skills that immigrants acquired prior to migration, I limit the analysis to those immigrants who entered the United States in the 5-year period just before the census cross-section. The sample consists of immigrant men aged 25–64, who worked at least one week in the year prior to the census, are not enrolled in school, and originated in one of the 80 largest sending countries.[21] An immigrant's weekly earnings is defined by the ratio of total earned income to annual weeks worked. The cross-country regression model is then given by:

$$\log \overline{w}_{kt} = \phi_t + Z_{kt}\alpha + \varepsilon, \tag{1.31}$$

where $\log \overline{w}_{kt}$ gives the mean log weekly wage of immigrants who originated in source country k and arrived just prior to the cross-section observed at time t; Z_{kt} is a vector of source-country characteristics; and ϕ_t is a vector of period fixed effects indicating the census of the observation.

Table 1.1 reports the coefficients from the cross-country regressions. The first regression specification uses the age-adjusted mean log weekly wage of the group as the dependent variable, while the second specification also adjusts for differences in educational attainment.[22] The model includes two variables summarizing the shape of the source country's income

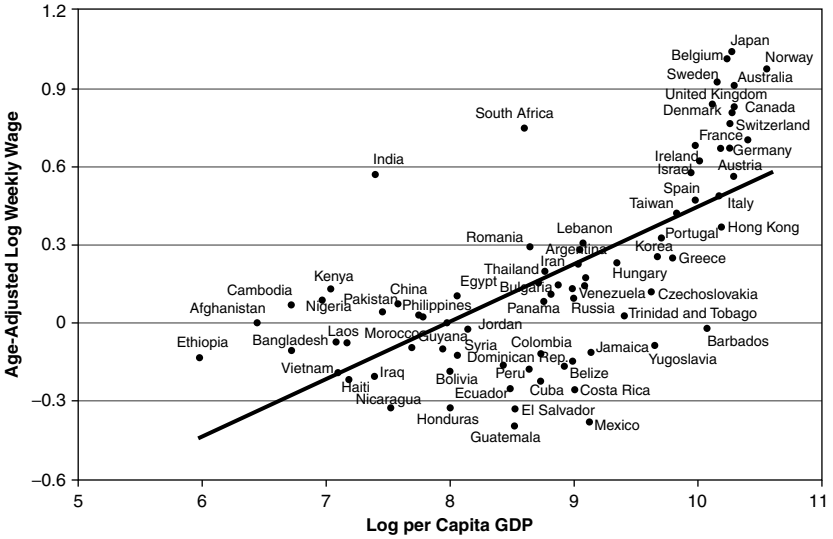
Table 1.1 Cross-country determinants of entry earnings of immigrants

	Specification	
Variable	(1)	(2)
Log per capita GDP in source country (PPP-adjusted)	0.116	0.115
	(0.030)	(0.026)
Gini coefficient in source country	−0.013	−0.008
	(0.005)	(0.004)
Distance from United States (=1 if distance exceeds 3,000 miles)	0.400	0.252
	(0.094)	(0.074)
Repressive regime (=1 if curtailed civil liberties)	−0.313	−0.266
	(0.058)	(0.048)
R^2	0.756	0.716
Immigrant earnings are adjusted for		
Age	Yes	Yes
Educational attainment	No	Yes

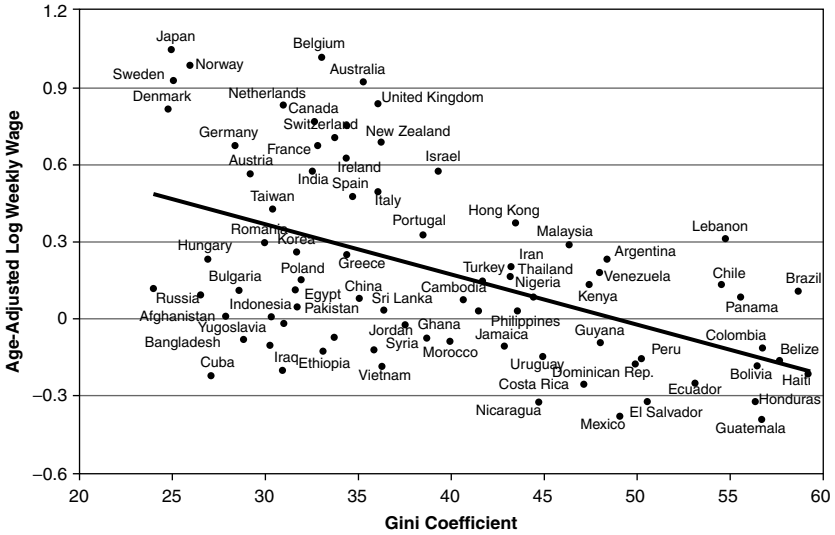
Sources: U.S. Census PUMS, 1980–2010; the Penn World Table (Heston et al. 2012); the World Development Indicators (World Bank 2013); and the Freedom in the World index (Freedom House 2011). See Appendix B for details.

Notes: The sample consists of immigrant men, aged 25–64, who originated in one of the 80 largest source countries, and who migrated to the United States in the 5-year period prior to each census between 1980 and 2010. The unit of observation is a country-year cell. Standard errors are reported in parentheses and are clustered at the country level. The dependent variable gives the adjusted log weekly wage of workers in the cell. The regressions are weighted by the sample size in each country-year cell, and include fixed effects indicating the decade in which the cohort entered the United States. The regressions have 320 observations. The adjusted log weekly wage in the cell is given by the country-of-origin fixed effect from individual-level regressions estimated separately in each census that control for age (introduced as a third-order polynomial) and years of education [in specification (2)].

distribution: the log per capita GDP and the Gini coefficient.[23] Both of these variables strongly influence immigrant outcomes: earnings rise with per capita GDP and fall with the Gini coefficient. As illustrated in Figure 1.4, these correlations are an important feature of the raw cross-country data. A doubling of the source country's per capita GDP increases the earnings of newly arrived immigrants in the United States by 12 percent, whereas a rise in the Gini coefficient from 36 to 49 (or the difference between the United Kingdom and Mexico) lowers earnings by about 17 percent.



A. Immigrant Earnings and Per Capita GDP



B. Immigrant Earnings and the Gini Coefficient

Figure 1.4. Source Country Determinants of Entry Earnings for 1995–1999 Immigrant Cohort

Source: U.S. Census PUMS, 2000; the Penn World Table (Heston et al. 2012), and the World Development Indicators (World Bank 2013). See Appendix B for details.

Notes: The construction of the age-adjusted log weekly wage for the national origin groups is described in the notes to Table 1.1. The *x*-axis gives either the (PPP-adjusted) log per capita GDP in the source country in 1995 (Panel A), or the average Gini coefficient in that country between 1978 and 2012 (Panel B).

The positive correlation between per capita income and immigrant earnings is sometimes interpreted as indicating that immigrants who originate in high-income countries find it easier to transfer their skills to other developed countries.[24] The negative correlation between immigrant earnings and the Gini coefficient is consistent with the Roy model prediction that immigrants who originate in countries with higher rates of return to skills should be, on average, relatively less skilled.[25]

The regression also includes a straightforward measure of "migration costs," defined by the distance between the United States and the source country. Immigrants originating in countries that are at least 3,000 miles away from the United States earn substantially more. Finally, the regression includes a civil liberties index constructed by Freedom House (2011).[26] Immigrants from countries with repressive regimes earn substantially less. Overall, the R^2 from these regressions is very high, so that a very small number of country-specific variables explain about 75 percent of the variation in mean entry earnings across national origin groups in the United States.

Grogger and Hanson (2011) use a variant of the cross-country empirical approach to estimate the (differenced) log odds ratio of migration implied by the linear utility model in equation (1.30). Specifically, they examine selection on educational attainment across a large number of sending and high-income receiving countries. The regression model compares the log odds ratio of migration for persons with a high level of education (13 or more years) with the log odds ratio for persons with a low-level of education (less than 9 years). As shown earlier, the linear utility model predicts a "generalized" form of positive selection. Skilled workers are more likely to migrate because the absolute wage gap between a poor and rich country widens at higher skill levels (even if the rate of return to skills were somewhat smaller in richer countries). The data confirm the positive correlation predicted by equation (1.30).

It is important to emphasize, however, that the empirical literature sometimes muddies the nature of the evidence by examining selection along different dimensions: selection in observables (such as educational attainment), selection in unobserved ability, and selection in the total value of the human capital stock. It is well known that education accounts for only a small portion of the variance in earnings across workers, suggesting that the nature of selection in educational attainment may not necessarily "transfer over" to a more comprehensive measure of a worker's

human capital.[27] I will return to this point shortly in the context of a specific immigrant flow.

Finally, it is well known that correlations estimated in cross-country regressions are contaminated by many extraneous factors (Levine and Renelt 1992). It is also difficult to ascertain which underlying forces any specific regressor in the cross-country regression truly represents. As an example, a high level of per capita GDP in the source country may represent both a high average income as well as the ease of transferring skills to another industrialized economy. As a result, much of the empirical work in recent years has shifted away from estimating cross-country correlations and instead chosen to analyze specific case studies of migration flows to determine if the flows are consistent with a particular selection model.[28]

Mexico

In the past decade, several international data sets have become available that permit more direct tests of the implications of the Roy model. These data often allow a straightforward comparison of the skills of two distinct groups of persons in a particular source country: those who choose to move and those who choose to stay. The pioneering study by Chiquiar and Hanson (2005) represents the first use of this approach in a context that is of particular interest in the United States, the selection of Mexican migrants. The number of Mexican-born persons residing in the United States increased at an unprecedented rate in recent decades; they now account for about a third of the foreign-born population. In fact, the number of Mexican immigrants as a fraction of the Mexican population rose from 1.5 percent in 1960 to 10.5 percent by 2005 (Hanson and McIntosh 2010).

The Chiquiar-Hanson study introduced the methodological "trick" of comparing the skill distributions of Mexican stayers and Mexican movers by measuring the skills of Mexican stayers in the sample enumerated in the Mexican census, and measuring the skills of Mexican movers in the sample of Mexican-born persons enumerated in the U.S. census.[29] In terms of the Roy model, a comparison of the wage distributions of Mexico and the United States suggests that the migrant flow should be negatively selected, because the rate of return to skills is much larger in Mexico. In 2000 the wage gap between workers with a high school diploma and workers with 9 to 11 years of schooling was 24.7 percent in Mexico, as compared to only 7.9 percent among Mexican immigrants in the United States,

and 10.2 percent among non-Mexican immigrants (Chiquiar and Hanson 2005, p. 256).

Table 1.2 summarizes the results of the Chiquiar-Hanson study by reporting the educational distribution in the samples of movers and stayers. There is no evidence that Mexican migrants to the United States are positively selected. In 2000, 11.3 percent of the Mexican "stayers" had at least 16 years of schooling as compared to only 5.0 percent of the immigrants. At the same time, however, the data indicate that Mexican migrants are not negatively selected: 69.4 percent of the Mexican stayers had less than 9 years of schooling, but only 60.1 percent of the Mexican immigrants did. Instead, the emigration rate is highest in the middle of the Mexican education distribution: 19.3 percent of the stayers have 10 to 15 years of schooling, as compared to 35.0 percent of the migrants.

Chiquiar and Hanson proposed a modified version of the Roy model that could fit the observed facts. In particular, it may be that the Roy model implication of negative selection is correct, but liquidity constraints and high mobility costs prevent the least-educated Mexicans from making the move. Even though the migrant sample is negatively selected, it is truncated from below because the high costs cut off the migration valve for the least skilled. The observed Mexican migrant sample would then be perceived as being characterized by "intermediate" selection.

An obvious problem with the Chiquiar-Hanson methodology is that the U.S. census does not provide a full count of the Mexican migrant population. A large fraction of the Mexican outflow is composed of

Table 1.2 Percent distribution of educational attainment for movers and stayers in Mexico, 2000

	Group			
Educational attainment	Mexican stayers	Mexican immigrants in the United States	Other immigrants in the United States	U.S. native born
0–9 years	69.4	60.1	14.4	4.5
10–15 years	19.3	35.0	46.2	68.9
16+ years	11.3	5.0	39.4	26.7

Source: Chiquiar and Hanson (2005, p. 248).

Notes: Columns may not add to 100 because of rounding error. All statistics refer to the distribution of educational attainment among men.

undocumented immigrants, and the U.S. census systematically undercounts both documented and undocumented immigrants, with the undercount rate being perhaps as high as 10 to 25 percent of Mexican migrants (Hanson 2006). It seems likely that the skill distribution of the "missing" Mexican immigrants differs systematically from that of the enumerated immigrants.

Fernández-Huertas (2011) revisited the issue by examining an alternative data set that is not contaminated by the undercount problem. The Encuesta Nacional de Empleo Trimestral (ENET) is a quarterly national labor force survey in Mexico that has a sampling structure similar to the Current Population Survey (CPS) in the United States. Specifically, the sample follows a particular household for five consecutive quarters. This implies that the sample contains observations on workers who at some point during their participation in the survey choose to migrate to the United States (legally or otherwise). As a result, the survey allows the researcher to observe the skills of the migrants *prior* to the move.

The use of the ENET also addresses an additional limitation of the census-based approach. The ENET reports not only the educational attainment of the sample of (eventual) migrants, but also their pre-migration earnings. Hence, the analysis can determine the nature of selection on both observable and unobservable determinants of productivity.

The empirical results from the ENET seem more consistent with the implications of the Roy model. The ENET survey respondents who eventually left Mexico have less education and lower earnings than the Mexican workers who chose to stay. The average Mexican migrant has 7.2 years of schooling, as compared to 8.5 years for the average stayer. Similarly, the average migrant earns about 29 percent less than the average stayer (Fernández-Huertas 2011, p. 76).

Figure 1.5 documents the negative selection in earnings by illustrating the probability distribution functions of the incomes of stayers and movers in the ENET. Recall that the Roy model implies that the earnings distribution of stayers will stochastically dominate that of movers if there is negative selection. Figure 1.5 confirms this implication of the theory, and a Kolmogorov-Smirnov test rejects the hypothesis that the two distributions are the same. However, it is important to emphasize that the ENET evidence may itself be problematic, because the sample is not representative of the Mexican population. By construction, the survey does not enumerate anyone belonging to households where all the members left Mexico

Figure 1.5. Negative Selection and Stochastic Dominance in Mexico

Source: Fernández-Huertas (2011, p. 79).

Note: The probability distribution functions are calculated using the sample of Mexican men working in the urban sector.

prior to the survey. Moreover, the ENET suffers from a high attrition rate: as many as 25 percent of the households do not survive the five quarters.

Kaestner and Malamud (2014) revisit the issue yet again using an alternative Mexico-based sample, the Mexican Family Life Survey (MxFLS), which is designed to be more representative than the ENET. Surprisingly, the MxFLS data suggest that Mexican migrants are intermediately selected in terms of educational attainment *and* negatively selected in terms of earnings. Specifically, Kaestner and Malamud document that the emigration rate for Mexican men with 7 to 9 years of education is higher than that of men with either more or less education. At the same time, however, the emigration rate is highest for workers in the lowest quintile of the income distribution and lowest for workers in the highest quintile.[30]

The Kaestner-Malamud study raises the interesting possibility that there could be different types of selection along different dimensions of skills. As noted earlier, evidence of positive or intermediate selection in terms of educational attainment may not accurately predict the selection that would be observed along other dimensions. We do not know, however,

why the relative rates of return to skills between any two countries (which presumably drive the differential types of selection) should differ so dramatically between observed and unobserved skills.[31]

Puerto Rico

The Jones Act of 1917 granted U.S. citizenship to all Puerto Ricans, implying that Puerto Ricans can move freely to the United States. As a result, Puerto Rico provides an opportunity to test the implications of selection models in the absence of any distortions arising from immigration policy restrictions.[32]

By 2000, around a third of the Puerto Rican–born population lived in the United States. In addition to this outflow, there is an additional flow in the other direction that is less well known. In particular, there has also been a sizable in-migration *of persons not born in Puerto Rico,* and the in-migrants account for about 10 percent of the Puerto Rican population. Interestingly, nearly two-thirds of the in-migrants were born in the United States but had Puerto Rican ancestry. The Puerto Rican context, therefore, represents a unique opportunity: there are sizable inflows and outflows, and the Roy model has very different predictions for the skill composition of these two groups.

The rate of return to skills is higher in Puerto Rico than in the United States. The age-adjusted log wage gap between college graduates and high school graduates in 1990 was 0.623 in the United States and 0.811 in Puerto Rico. A straightforward application of the Roy model suggests a simple sorting: the least-skilled Puerto Rican–born workers have the greatest incentive to leave Puerto Rico, and Puerto Rico should attract high-skill workers from abroad.

It is easy to test these implications by manipulating the available U.S. and Puerto Rican census data to define the groups of movers and stayers (in the spirit of the Chiquiar-Hanson study). Consider first the migration flow from Puerto Rico to the United States. In this flow, the movers are persons born in Puerto Rico but living in the United States, whereas the stayers are persons born in Puerto Rico and still living in Puerto Rico. The data reported in Table 1.3 document the differences in the distribution of educational attainment between these two groups. It is evident that movers are more likely to be high school dropouts and less likely to be college graduates. In particular, 19.7 percent of the stayers are high school dropouts,

Table 1.3 Percent distribution of educational attainment for movers and stayers in Puerto Rico, 2000

	1. Migration from Puerto Rico to the United States		2. Migration from the United States to Puerto Rico	
Educational attainment	Movers	Stayers	Movers	Stayers
0–11 years	26.2	19.7	10.8	18.0
12 years	35.7	33.6	29.9	38.6
13–15 years	23.8	26.1	34.9	30.6
16+ years	14.3	20.6	24.4	12.7

Source: Adapted from Borjas (2008, pp. 41–45).

Notes: Columns may not add to 100 because of rounding error. All statistics refer to the distribution of educational attainment among men.

but this education group composes over a quarter of the movers. Similarly, 20.6 percent of the stayers are college graduates, but only 14.3 percent of the movers are.

Consider instead the migration flow from the United States to Puerto Rico. In this flow, the movers are persons born in the United States of Puerto Rican ancestry, but living in Puerto Rico; while the stayers are persons born in the United States of Puerto Rican ancestry and still living in the United States. Interestingly, the skill distributions of the movers and stayers in this reverse flow are a mirror image of the Puerto Rico–to-U.S. outflow. The movers are more likely to be college graduates and less likely to be high school dropouts. In short, as implied by the Roy model, the evidence clearly indicates that the type of worker leaving Puerto Rico differs dramatically from the type of worker moving in.

Norway, 1900

Although most of the existing studies use international migration flows in the modern era to test the implications of selection models, the recent availability of historical microdata sets has sparked interest in examining the selection of immigrants during the Great Migration of 1880–1920. At that time the United States imposed few restrictions limiting immigration from many source countries, so the analysis, as in the Puerto Rican context, allows a "clean" test of the implications of income maximization. A

recent study by Abramitzky et al. (2012) shows the value and promise of analyzing these types of historical data sets.

During the (First) Great Migration, Norway had one of the highest out-migration rates in Europe; over a quarter of its population migrated to the United States. Abramitzky et al. exploited the fact that Norway has now digitized its two censuses from that period (1865 and 1890), and merged these data with newly available genealogical records in the United States (drawn from decennial censuses at the time) that contain information on all Norwegian-born men living in the United States.

Norway had a much more unequal income distribution than the United States at the beginning of the twentieth century: "United States workers below the fiftieth percentile of the earnings distribution out-earned similar Norwegians, while Norwegians above the ninetieth percentile commanded higher earnings than their U.S. counterparts" (Abramitzky et al. 2012, p. 1837). The Roy model would then imply that Norwegian immigrants to the United States should be negatively selected.

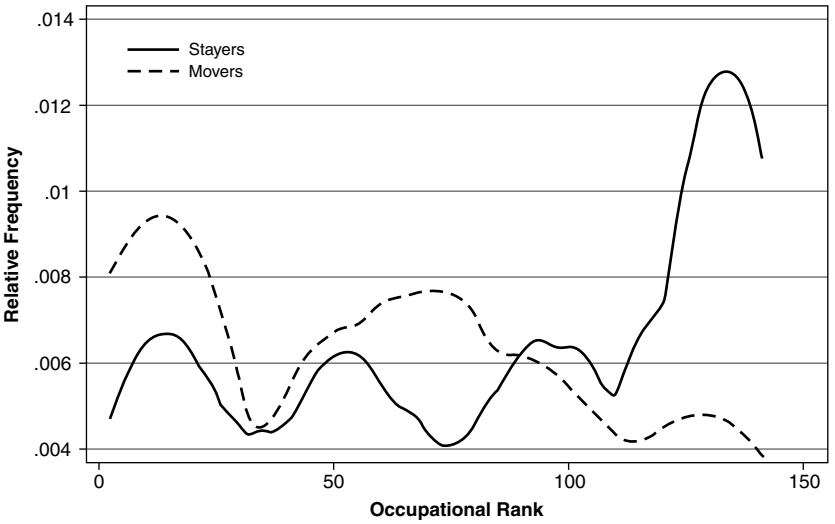


Figure 1.6. Selection of Migrants from Norway to the United States, circa 1900

Source: Abramitzky et al. (2012, p. 1843).

Note: Occupations are ranked from lowest to highest earnings. The figure gives the fraction of Norwegian men born in urban areas who are employed at each occupational rank.

The data in the various censuses report a worker's occupation in Norway (for the stayers) or in the United States (for the movers). The skills required by each occupation can be summarized by a measure of "occupational rank" (that is, the mean earnings that workers in a particular occupation earned in the United States in 1901). Figure 1.6 illustrates the nature of the selection by showing the fraction of either movers or stayers employed at a particular point in the occupational earnings distribution. Norwegian migrants in the United States are clearly more likely to be employed in low-paying occupations than the Norwegians who chose to remain in Norway, and less likely to be employed in high-paying occupations. The evidence, therefore, seems consistent with a simple version of the Roy model where high-skill Norwegians chose to remain in a country that, at the time, offered a relatively higher payoff for their abilities.